by_normal_form_q17 over GF(17)

June 30, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + 4X_1^4 + 2X_2^4 + 15X_0^3X_1 + 6X_0^3X_2 + 6X_0X_1^3 + 14X_1^3X_2 + 6X_0X_2^3$$
$$+8X_1X_2^3 + 12X_0^2X_1^2 + 15X_0^2X_2^2 + 15X_1^2X_2^2 + 3X_0^2X_1X_2 + 15X_0X_1^2X_2$$

$$(1, 4, 2, 15, 6, 6, 14, 6, 8, 12, 15, 15, 3, 15, 0)$$

```
X0^4 + 4*X1^4 + 2*X2^4 + 15*X0^3*X1 + 6*X0^3*X2 + 6*X0*X1^3 + 14*X1^3*X2 + 6*X0*X2^3 + 8*X1*
```

General information

Number of bitangents	28
Number of points	12
Fullness	is full
Number of Kovalevski points	15
Bitangent line type (a_0, a_1, a_2)	$(16, 12, 0)$
Number of singular points	0

All points on the curve

The quartic has 12 points:
The points on the quartic curve are:

$0: P_{28} = (9, 0, 1)$
$1: P_{92} = (6, 4, 1)$
$2: P_{123} = (3, 6, 1)$
$3: P_{135} = (15, 6, 1)$
$4: P_{138} = (1, 7, 1)$
$5: P_{144} = (7, 7, 1)$
$6: P_{161} = (7, 8, 1)$
$7: P_{217} = (12, 11, 1)$
$8: P_{227} = (5, 12, 1)$
$9: P_{232} = (10, 12, 1)$
$10: P_{253} = (14, 13, 1)$
$11: P_{305} = (15, 16, 1)$

The points by rank are: (28, 92, 123, 135, 138, 144, 161, 217, 227, 232, 253, 305)

The Kovalevski points are:
$0: P_{127} = (7,6,1) = b_1 \cap b_6 \cap c_{14} \cap c_{46}$
$1: P_{256} = (0,14,1) = a_3 \cap a_4 \cap c_{36} \cap c_{46}$
$2: P_{116} = (13,5,1) = a_4 \cap b_5 \cap c_{45} \cap d$
$3: P_{230} = (8,12,1) = a_2 \cap b_6 \cap c_{26} \cap d$
$4: P_{204} = (16,10,1) = c_{12} \cap c_{14} \cap c_{25} \cap c_{45}$
$5: P_{190} = (2,10,1) = a_4 \cap b_6 \cap c_{12} \cap c_{35}$
$6: P_{188} = (0,10,1) = a_2 \cap a_3 \cap c_{12} \cap c_{13}$
$7: P_{186} = (15,9,1) = c_{13} \cap c_{25} \cap c_{46} \cap d$
$8: P_{77} = (8,3,1) = a_1 \cap a_5 \cap c_{16} \cap c_{56}$
$9: P_{299} = (9,16,1) = b_2 \cap b_3 \cap c_{24} \cap c_{34}$
$10: P_{36} = (0,1,1) = a_3 \cap b_5 \cap c_{14} \cap c_{26}$
$11: P_{148} = (11,7,1) = b_1 \cap b_5 \cap c_{13} \cap c_{35}$
$12: P_{277} = (4,15,1) = c_{25} \cap c_{26} \cap c_{35} \cap c_{36}$
$13: P_{66} = (14,2,1) = a_6 \cap b_4 \cap c_{15} \cap c_{23}$
$14: P_7 = (4,1,0) = a_2 \cap b_1 \cap c_{36} \cap c_{45}$
The incidence structure induced by the Kovalevski points is:
Flags=(8, 18, 21, 29, 31, 36, 40, 46, 47, 50, 68, 88, 90, 101, 104, 114, 129, 148, 152, 160, 161, 165, 168, 170, 184, 185, 186, 201, 202, 206, 210, 214, 220, 238, 248, 268, 279, 289, 292, 297, 303, 310, 312, 324, 335, 341, 342, 346, 357, 359, 362, 364, 374, 375, 376, 382, 398, 407, 408, 412)
nb_rows=28
nb_cols=15
nb_flags=60
Incma=
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1)
(0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0)
(1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0)
(1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0)
(0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0)

(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0)
(0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1)
(0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
(1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0)
(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)
(0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0)
The Kovalevski points sorted by rank are: (7, 36, 66, 77, 116, 127, 148, 186, 188, 190, 204, 230, 256, 277, 299)
The points off the curve are:

$0: P_0 = (1,0,0)$
$1: P_1 = (0,1,0)$
$2: P_2 = (0,0,1)$
$3: P_3 = (1,1,1)$
$4: P_4 = (1,1,0)$
$5: P_5 = (2,1,0)$
$6: P_6 = (3,1,0)$
$7: P_7 = (4,1,0)$
$8: P_8 = (5,1,0)$
$9: P_9 = (6,1,0)$
$10: P_{10} = (7,1,0)$
$11: P_{11} = (8,1,0)$
$12: P_{12} = (9,1,0)$
$13: P_{13} = (10,1,0)$
$14: P_{14} = (11,1,0)$
$15: P_{15} = (12,1,0)$
$16: P_{16} = (13,1,0)$
$17: P_{17} = (14,1,0)$
$18: P_{18} = (15,1,0)$
$19: P_{19} = (16,1,0)$
$20: P_{20} = (1,0,1)$
$21: P_{21} = (2,0,1)$
$22: P_{22} = (3,0,1)$
$23: P_{23} = (4,0,1)$
$24: P_{24} = (5,0,1)$
$25: P_{25} = (6,0,1)$
$26: P_{26} = (7,0,1)$
$27: P_{27} = (8,0,1)$
$28: P_{29} = (10,0,1)$
$29: P_{30} = (11,0,1)$
$30: P_{31} = (12,0,1)$
$31: P_{32} = (13,0,1)$
$32: P_{33} = (14,0,1)$
$33: P_{34} = (15,0,1)$
$34: P_{35} = (16,0,1)$
$35: P_{36} = (0,1,1)$
$36: P_{37} = (2,1,1)$
$37: P_{38} = (3,1,1)$
$38: P_{39} = (4,1,1)$
$39: P_{40} = (5,1,1)$
$40: P_{41} = (6,1,1)$
$41: P_{42} = (7,1,1)$
$42: P_{43} = (8,1,1)$
$43: P_{44} = (9,1,1)$
$44: P_{45} = (10,1,1)$
$45: P_{46} = (11,1,1)$
$46: P_{47} = (12,1,1)$
$47: P_{48} = (13,1,1)$
$48: P_{49} = (14,1,1)$
$49: P_{50} = (15,1,1)$
$50: P_{51} = (16,1,1)$
$51: P_{52} = (0,2,1)$
$52: P_{53} = (1,2,1)$
$53: P_{54} = (2,2,1)$
$54: P_{55} = (3,2,1)$
$55: P_{56} = (4,2,1)$
$56: P_{57} = (5,2,1)$
$57: P_{58} = (6,2,1)$
$58: P_{59} = (7,2,1)$
$59: P_{60} = (8,2,1)$
$60: P_{61} = (9,2,1)$
$61: P_{62} = (10,2,1)$
$62: P_{63} = (11,2,1)$
$63: P_{64} = (12,2,1)$
$64: P_{65} = (13,2,1)$
$65: P_{66} = (14,2,1)$
$66: P_{67} = (15,2,1)$
$67: P_{68} = (16,2,1)$
$68: P_{69} = (0,3,1)$
$69: P_{70} = (1,3,1)$
$70: P_{71} = (2,3,1)$
$71: P_{72} = (3,3,1)$
$72: P_{73} = (4,3,1)$
$73: P_{74} = (5,3,1)$
$74: P_{75} = (6,3,1)$
$75: P_{76} = (7,3,1)$
$76: P_{77} = (8,3,1)$
$77: P_{78} = (9,3,1)$
$78: P_{79} = (10,3,1)$
$79: P_{80} = (11,3,1)$
$80: P_{81} = (12,3,1)$
$81: P_{82} = (13,3,1)$
$82: P_{83} = (14,3,1)$
$83: P_{84} = (15,3,1)$
$84: P_{85} = (16,3,1)$
$85: P_{86} = (0,4,1)$
$86: P_{87} = (1,4,1)$
$87: P_{88} = (2,4,1)$
$88: P_{89} = (3,4,1)$
$89: P_{90} = (4,4,1)$
$90: P_{91} = (5,4,1)$
$91: P_{93} = (7,4,1)$
$92: P_{94} = (8,4,1)$
$93: P_{95} = (9,4,1)$
$94: P_{96} = (10,4,1)$
$95: P_{97} = (11,4,1)$
$96: P_{98} = (12,4,1)$
$97: P_{99} = (13,4,1)$
$98: P_{100} = (14,4,1)$
$99: P_{101} = (15,4,1)$
$100: P_{102} = (16,4,1)$
$101: P_{103} = (0,5,1)$

$102: P_{104} = (1, 5, 1)$
$103: P_{105} = (2, 5, 1)$
$104: P_{106} = (3, 5, 1)$
$105: P_{107} = (4, 5, 1)$
$106: P_{108} = (5, 5, 1)$
$107: P_{109} = (6, 5, 1)$
$108: P_{110} = (7, 5, 1)$
$109: P_{111} = (8, 5, 1)$
$110: P_{112} = (9, 5, 1)$
$111: P_{113} = (10, 5, 1)$
$112: P_{114} = (11, 5, 1)$
$113: P_{115} = (12, 5, 1)$
$114: P_{116} = (13, 5, 1)$
$115: P_{117} = (14, 5, 1)$
$116: P_{118} = (15, 5, 1)$
$117: P_{119} = (16, 5, 1)$
$118: P_{120} = (0, 6, 1)$
$119: P_{121} = (1, 6, 1)$
$120: P_{122} = (2, 6, 1)$
$121: P_{124} = (4, 6, 1)$
$122: P_{125} = (5, 6, 1)$
$123: P_{126} = (6, 6, 1)$
$124: P_{127} = (7, 6, 1)$
$125: P_{128} = (8, 6, 1)$
$126: P_{129} = (9, 6, 1)$
$127: P_{130} = (10, 6, 1)$
$128: P_{131} = (11, 6, 1)$
$129: P_{132} = (12, 6, 1)$
$130: P_{133} = (13, 6, 1)$
$131: P_{134} = (14, 6, 1)$
$132: P_{136} = (16, 6, 1)$
$133: P_{137} = (0, 7, 1)$
$134: P_{139} = (2, 7, 1)$
$135: P_{140} = (3, 7, 1)$
$136: P_{141} = (4, 7, 1)$
$137: P_{142} = (5, 7, 1)$
$138: P_{143} = (6, 7, 1)$
$139: P_{145} = (8, 7, 1)$
$140: P_{146} = (9, 7, 1)$
$141: P_{147} = (10, 7, 1)$
$142: P_{148} = (11, 7, 1)$
$143: P_{149} = (12, 7, 1)$
$144: P_{150} = (13, 7, 1)$
$145: P_{151} = (14, 7, 1)$
$146: P_{152} = (15, 7, 1)$
$147: P_{153} = (16, 7, 1)$
$148: P_{154} = (0, 8, 1)$
$149: P_{155} = (1, 8, 1)$
$150: P_{156} = (2, 8, 1)$
$151: P_{157} = (3, 8, 1)$
$152: P_{158} = (4, 8, 1)$
$153: P_{159} = (5, 8, 1)$
$154: P_{160} = (6, 8, 1)$
$155: P_{162} = (8, 8, 1)$
$156: P_{163} = (9, 8, 1)$
$157: P_{164} = (10, 8, 1)$
$158: P_{165} = (11, 8, 1)$
$159: P_{166} = (12, 8, 1)$
$160: P_{167} = (13, 8, 1)$
$161: P_{168} = (14, 8, 1)$
$162: P_{169} = (15, 8, 1)$
$163: P_{170} = (16, 8, 1)$
$164: P_{171} = (0, 9, 1)$
$165: P_{172} = (1, 9, 1)$
$166: P_{173} = (2, 9, 1)$
$167: P_{174} = (3, 9, 1)$
$168: P_{175} = (4, 9, 1)$
$169: P_{176} = (5, 9, 1)$
$170: P_{177} = (6, 9, 1)$
$171: P_{178} = (7, 9, 1)$
$172: P_{179} = (8, 9, 1)$
$173: P_{180} = (9, 9, 1)$
$174: P_{181} = (10, 9, 1)$
$175: P_{182} = (11, 9, 1)$
$176: P_{183} = (12, 9, 1)$
$177: P_{184} = (13, 9, 1)$
$178: P_{185} = (14, 9, 1)$
$179: P_{186} = (15, 9, 1)$
$180: P_{187} = (16, 9, 1)$
$181: P_{188} = (0, 10, 1)$
$182: P_{189} = (1, 10, 1)$
$183: P_{190} = (2, 10, 1)$
$184: P_{191} = (3, 10, 1)$
$185: P_{192} = (4, 10, 1)$
$186: P_{193} = (5, 10, 1)$
$187: P_{194} = (6, 10, 1)$
$188: P_{195} = (7, 10, 1)$
$189: P_{196} = (8, 10, 1)$
$190: P_{197} = (9, 10, 1)$
$191: P_{198} = (10, 10, 1)$
$192: P_{199} = (11, 10, 1)$
$193: P_{200} = (12, 10, 1)$
$194: P_{201} = (13, 10, 1)$
$195: P_{202} = (14, 10, 1)$
$196: P_{203} = (15, 10, 1)$
$197: P_{204} = (16, 10, 1)$
$198: P_{205} = (0, 11, 1)$
$199: P_{206} = (1, 11, 1)$
$200: P_{207} = (2, 11, 1)$
$201: P_{208} = (3, 11, 1)$
$202: P_{209} = (4, 11, 1)$
$203: P_{210} = (5, 11, 1)$
$204: P_{211} = (6, 11, 1)$
$205: P_{212} = (7, 11, 1)$
$206: P_{213} = (8, 11, 1)$
$207: P_{214} = (9, 11, 1)$
$208: P_{215} = (10, 11, 1)$
$209: P_{216} = (11, 11, 1)$
$210: P_{218} = (13, 11, 1)$
$211: P_{219} = (14, 11, 1)$
$212: P_{220} = (15, 11, 1)$
$213: P_{221} = (16, 11, 1)$
$214: P_{222} = (0, 12, 1)$
$215: P_{223} = (1, 12, 1)$
$216: P_{224} = (2, 12, 1)$
$217: P_{225} = (3, 12, 1)$
$218: P_{226} = (4, 12, 1)$
$219: P_{228} = (6, 12, 1)$
$220: P_{229} = (7, 12, 1)$
$221: P_{230} = (8, 12, 1)$
$222: P_{231} = (9, 12, 1)$
$223: P_{233} = (11, 12, 1)$
$224: P_{234} = (12, 12, 1)$
$225: P_{235} = (13, 12, 1)$
$226: P_{236} = (14, 12, 1)$
$227: P_{237} = (15, 12, 1)$
$228: P_{238} = (16, 12, 1)$
$229: P_{239} = (0, 13, 1)$
$230: P_{240} = (1, 13, 1)$
$231: P_{241} = (2, 13, 1)$
$232: P_{242} = (3, 13, 1)$
$233: P_{243} = (4, 13, 1)$
$234: P_{244} = (5, 13, 1)$
$235: P_{245} = (6, 13, 1)$
$236: P_{246} = (7, 13, 1)$
$237: P_{247} = (8, 13, 1)$
$238: P_{248} = (9, 13, 1)$
$239: P_{249} = (10, 13, 1)$

$240: P_{250} = (11, 13, 1)$
$241: P_{251} = (12, 13, 1)$
$242: P_{252} = (13, 13, 1)$
$243: P_{254} = (15, 13, 1)$
$244: P_{255} = (16, 13, 1)$
$245: P_{256} = (0, 14, 1)$
$246: P_{257} = (1, 14, 1)$
$247: P_{258} = (2, 14, 1)$
$248: P_{259} = (3, 14, 1)$
$249: P_{260} = (4, 14, 1)$
$250: P_{261} = (5, 14, 1)$
$251: P_{262} = (6, 14, 1)$
$252: P_{263} = (7, 14, 1)$
$253: P_{264} = (8, 14, 1)$
$254: P_{265} = (9, 14, 1)$
$255: P_{266} = (10, 14, 1)$
$256: P_{267} = (11, 14, 1)$
$257: P_{268} = (12, 14, 1)$
$258: P_{269} = (13, 14, 1)$
$259: P_{270} = (14, 14, 1)$
$260: P_{271} = (15, 14, 1)$
$261: P_{272} = (16, 14, 1)$
$262: P_{273} = (0, 15, 1)$
$263: P_{274} = (1, 15, 1)$
$264: P_{275} = (2, 15, 1)$
$265: P_{276} = (3, 15, 1)$
$266: P_{277} = (4, 15, 1)$
$267: P_{278} = (5, 15, 1)$
$268: P_{279} = (6, 15, 1)$
$269: P_{280} = (7, 15, 1)$
$270: P_{281} = (8, 15, 1)$
$271: P_{282} = (9, 15, 1)$
$272: P_{283} = (10, 15, 1)$
$273: P_{284} = (11, 15, 1)$
$274: P_{285} = (12, 15, 1)$
$275: P_{286} = (13, 15, 1)$
$276: P_{287} = (14, 15, 1)$
$277: P_{288} = (15, 15, 1)$
$278: P_{289} = (16, 15, 1)$
$279: P_{290} = (0, 16, 1)$
$280: P_{291} = (1, 16, 1)$
$281: P_{292} = (2, 16, 1)$
$282: P_{293} = (3, 16, 1)$
$283: P_{294} = (4, 16, 1)$
$284: P_{295} = (5, 16, 1)$
$285: P_{296} = (6, 16, 1)$
$286: P_{297} = (7, 16, 1)$
$287: P_{298} = (8, 16, 1)$
$288: P_{299} = (9, 16, 1)$
$289: P_{300} = (10, 16, 1)$
$290: P_{301} = (11, 16, 1)$
$291: P_{302} = (12, 16, 1)$
$292: P_{303} = (13, 16, 1)$
$293: P_{304} = (14, 16, 1)$
$294: P_{306} = (16, 16, 1)$

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 136, 137, 139, 140, 141, 142, 143, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 218, 219, 220, 221, 222, 223, 224, 225, 226, 228, 229, 230, 231, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 306)

Duals of Bitangents

dual of bitangents:
$0: P_{95} = (9, 4, 1)$
$1: P_{106} = (3, 5, 1)$
$2: P_{0} = (1, 0, 0)$
$3: P_{132} = (12, 6, 1)$

$4 : P_{170} = (16, 8, 1)$
$5 : P_{25} = (6, 0, 1)$
$6 : P_7 = (4, 1, 0)$
$7 : P_{36} = (0, 1, 1)$
$8 : P_{54} = (2, 2, 1)$
$9 : P_{207} = (2, 11, 1)$
$10 : P_{306} = (16, 16, 1)$
$11 : P_{185} = (14, 9, 1)$
$12 : P_{103} = (0, 5, 1)$
$13 : P_{109} = (6, 5, 1)$
$14 : P_{298} = (8, 16, 1)$
$15 : P_{142} = (5, 7, 1)$
$16 : P_{228} = (6, 12, 1)$
$17 : P_{89} = (3, 4, 1)$
$18 : P_{227} = (5, 12, 1)$
$19 : P_{83} = (14, 3, 1)$
$20 : P_{302} = (12, 16, 1)$
$21 : P_{34} = (15, 0, 1)$
$22 : P_{197} = (9, 10, 1)$
$23 : P_{127} = (7, 6, 1)$
$24 : P_{261} = (5, 14, 1)$
$25 : P_{122} = (2, 6, 1)$
$26 : P_{280} = (7, 15, 1)$
$27 : P_{247} = (8, 13, 1)$
Dimension of the vanishing ideal is 0

The lines and their points of contact are:
$a_1 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 13 \end{bmatrix}_{157} P_{217} = \mathbf{P}(12, 11, 1)\ 4\times$
$a_2 = \begin{bmatrix} 1 & 0 & 14 \\ 0 & 1 & 12 \end{bmatrix}_{264}$
$a_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{306}$
$a_4 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 11 \end{bmatrix}_{101}$
$a_5 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 9 \end{bmatrix}_{27} P_{135} = \mathbf{P}(15, 6, 1)\ 4\times$
$a_6 = \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 0 \end{bmatrix}_{198} P_{253} = \mathbf{P}(14, 13, 1)\ 4\times$
$b_1 = \begin{bmatrix} 1 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{251}$
$b_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 16 \end{bmatrix}_{16} P_{305} = \mathbf{P}(15, 16, 1)\ 4\times$

$b_3 = \left[\begin{array}{ccc} 1 & 0 & 15 \\ 0 & 1 & 15 \end{array} \right]_{285}$ $P_{138} = \mathbf{P}(1,7,1)\ 4\times$

$b_4 = \left[\begin{array}{ccc} 1 & 0 & 15 \\ 0 & 1 & 6 \end{array} \right]_{276}$ $P_{232} = \mathbf{P}(10,12,1)\ 4\times$

$b_5 = \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right]_{19}$

$b_6 = \left[\begin{array}{ccc} 1 & 0 & 3 \\ 0 & 1 & 8 \end{array} \right]_{62}$

$c_{12} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 12 \end{array} \right]_{12}$

$c_{13} = \left[\begin{array}{ccc} 1 & 0 & 11 \\ 0 & 1 & 12 \end{array} \right]_{210}$

$c_{14} = \left[\begin{array}{ccc} 1 & 0 & 9 \\ 0 & 1 & 1 \end{array} \right]_{163}$

$c_{15} = \left[\begin{array}{ccc} 1 & 0 & 12 \\ 0 & 1 & 10 \end{array} \right]_{226}$ $P_{144} = \mathbf{P}(7,7,1)\ 4\times$

$c_{16} = \left[\begin{array}{ccc} 1 & 0 & 11 \\ 0 & 1 & 5 \end{array} \right]_{203}$ $P_{92} = \mathbf{P}(6,4,1)\ 4\times$

$c_{23} = \left[\begin{array}{ccc} 1 & 0 & 14 \\ 0 & 1 & 13 \end{array} \right]_{265}$ $P_{123} = \mathbf{P}(3,6,1)\ 4\times$

$c_{24} = \left[\begin{array}{ccc} 1 & 0 & 12 \\ 0 & 1 & 5 \end{array} \right]_{221}$ $P_{227} = \mathbf{P}(5,12,1)\ 4\times$

$c_{25} = \left[\begin{array}{ccc} 1 & 0 & 3 \\ 0 & 1 & 14 \end{array} \right]_{68}$

$c_{26} = \left[\begin{array}{ccc} 1 & 0 & 5 \\ 0 & 1 & 1 \end{array} \right]_{91}$

$c_{34} = \left[\begin{array}{ccc} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array} \right]_{36}$ $P_{28} = \mathbf{P}(9,0,1)\ 4\times$

$c_{35} = \left[\begin{array}{ccc} 1 & 0 & 8 \\ 0 & 1 & 7 \end{array} \right]_{151}$

$c_{36} = \left[\begin{array}{ccc} 1 & 0 & 10 \\ 0 & 1 & 11 \end{array} \right]_{191}$

$c_{45} = \left[\begin{array}{ccc} 1 & 0 & 12 \\ 0 & 1 & 3 \end{array} \right]_{219}$

$c_{46} = \left[\begin{array}{ccc} 1 & 0 & 15 \\ 0 & 1 & 11 \end{array} \right]_{281}$

$c_{56} = \left[\begin{array}{ccc} 1 & 0 & 10 \\ 0 & 1 & 2 \end{array} \right]_{182}$ $P_{161} = \mathbf{P}(7,8,1)\ 4\times$

$d = \left[\begin{array}{ccc} 1 & 0 & 9 \\ 0 & 1 & 4 \end{array} \right]_{166}$

Rank of lines: (157, 264, 306, 101, 27, 198, 251, 16, 285, 276, 19, 62, 12, 210, 163, 226, 203, 265, 221, 68, 91, 36, 151, 191, 219, 281, 182, 166)
Line type: 1^{12}, 0^{16}

12	1	26, 5, 21, 9, 4, 18, 17, 8, 16, 15, 7, 0
16	0	19, 12, 23, 20, 25, 6, 27, 24, 22, 10, 11, 2, 13, 14, 3, 1

point types: 1^{12}

12	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 0

point types for points off the curve: 4^{15}, 3^{48}, 2^{144}, 0^{88}

15	4	124, 245, 114, 221, 197, 183, 181, 179, 76, 288, 35, 142, 266, 65, 7
48	3	126, 242, 120, 235, 234, 28, 27, 225, 224, 13, 110, 211, 104, 207, 202, 98, 194, 190, 46, 184, 45, 180, 175, 43, 86, 85, 41, 162, 161, 156, 293, 145, 289, 287, 285, 284, 281, 280, 276, 271, 270, 133, 32, 264, 261, 256, 1, 294
144	2	141, 248, 30, 29, 241, 239, 119, 240, 231, 115, 229, 282, 59, 238, 116, 12, 108, 215, 107, 53, 219, 26, 111, 6, 213, 230, 232, 57, 105, 118, 52, 25, 121, 60, 208, 61, 252, 253, 51, 2, 122, 205, 101, 50, 201, 49, 200, 48, 97, 5, 193, 192, 191, 95, 11, 249, 189, 93, 254, 186, 185, 21, 89, 88, 176, 4, 143, 173, 84, 83, 20, 260, 166, 165, 82, 164, 81, 39, 80, 19, 169, 160, 157, 77, 155, 38, 292, 154, 177, 178, 151, 75, 90, 182, 18, 150, 149, 74, 187, 188, 148, 94, 73, 291, 144, 71, 195, 196, 17, 198, 132, 283, 140, 100, 203, 279, 138, 68, 0, 136, 272, 67, 33, 212, 268, 214, 66, 265, 131, 218, 263, 220, 54, 222, 262, 259, 112, 226, 129, 290, 63, 286, 137, 250
88	0	278, 258, 172, 72, 128, 91, 204, 64, 158, 147, 228, 31, 275, 8, 217, 102, 255, 174, 40, 159, 153, 36, 274, 55, 117, 237, 134, 24, 233, 62, 56, 109, 209, 206, 199, 70, 22, 87, 171, 168, 146, 79, 44, 37, 152, 96, 267, 135, 106, 69, 34, 236, 113, 58, 14, 277, 216, 123, 273, 139, 47, 3, 269, 243, 227, 223, 246, 210, 247, 103, 16, 99, 23, 244, 127, 92, 10, 257, 9, 251, 170, 167, 163, 15, 42, 130, 125, 78

Lines on points off the curve:
Off point 0 $= P_0 = (1, 0, 0)$ lies on 2 bisecants : { 7, 12 }

Off point 1 = $P_1 = (0, 1, 0)$ lies on 3 bisecants : { 2, 5, 21 }
Off point 2 = $P_2 = (0, 0, 1)$ lies on 2 bisecants : { 2, 6 }
Off point 3 = $P_3 = (1, 1, 1)$ lies on 0 bisecants : { }
Off point 4 = $P_4 = (1, 1, 0)$ lies on 2 bisecants : { 18, 19 }
Off point 5 = $P_5 = (2, 1, 0)$ lies on 2 bisecants : { 13, 15 }
Off point 6 = $P_6 = (3, 1, 0)$ lies on 2 bisecants : { 9, 11 }
Off point 7 = $P_7 = (4, 1, 0)$ lies on 4 bisecants : { 1, 6, 23, 24 }
Off point 8 = $P_8 = (5, 1, 0)$ lies on 0 bisecants : { }
Off point 9 = $P_9 = (6, 1, 0)$ lies on 0 bisecants : { }
Off point 10 = $P_{10} = (7, 1, 0)$ lies on 0 bisecants : { }
Off point 11 = $P_{11} = (8, 1, 0)$ lies on 2 bisecants : { 3, 4 }
Off point 12 = $P_{12} = (9, 1, 0)$ lies on 2 bisecants : { 0, 27 }
Off point 13 = $P_{13} = (10, 1, 0)$ lies on 3 bisecants : { 17, 20, 26 }
Off point 14 = $P_{14} = (11, 1, 0)$ lies on 0 bisecants : { }
Off point 15 = $P_{15} = (12, 1, 0)$ lies on 0 bisecants : { }
Off point 16 = $P_{16} = (13, 1, 0)$ lies on 0 bisecants : { }
Off point 17 = $P_{17} = (14, 1, 0)$ lies on 2 bisecants : { 22, 25 }
Off point 18 = $P_{18} = (15, 1, 0)$ lies on 2 bisecants : { 14, 16 }
Off point 19 = $P_{19} = (16, 1, 0)$ lies on 2 bisecants : { 8, 10 }
Off point 20 = $P_{20} = (1, 0, 1)$ lies on 2 bisecants : { 4, 10 }
Off point 21 = $P_{21} = (2, 0, 1)$ lies on 2 bisecants : { 14, 27 }
Off point 22 = $P_{22} = (3, 0, 1)$ lies on 0 bisecants : { }
Off point 23 = $P_{23} = (4, 0, 1)$ lies on 0 bisecants : { }
Off point 24 = $P_{24} = (5, 0, 1)$ lies on 0 bisecants : { }
Off point 25 = $P_{25} = (6, 0, 1)$ lies on 2 bisecants : { 11, 19 }
Off point 26 = $P_{26} = (7, 0, 1)$ lies on 2 bisecants : { 3, 20 }
Off point 27 = $P_{27} = (8, 0, 1)$ lies on 3 bisecants : { 8, 9, 25 }
Off point 28 = $P_{29} = (10, 0, 1)$ lies on 3 bisecants : { 15, 18, 24 }
Off point 29 = $P_{30} = (11, 0, 1)$ lies on 2 bisecants : { 1, 17 }
Off point 30 = $P_{31} = (12, 0, 1)$ lies on 2 bisecants : { 23, 26 }
Off point 31 = $P_{32} = (13, 0, 1)$ lies on 0 bisecants : { }
Off point 32 = $P_{33} = (14, 0, 1)$ lies on 3 bisecants : { 5, 13, 16 }
Off point 33 = $P_{34} = (15, 0, 1)$ lies on 2 bisecants : { 0, 22 }
Off point 34 = $P_{35} = (16, 0, 1)$ lies on 0 bisecants : { }
Off point 35 = $P_{36} = (0, 1, 1)$ lies on 4 bisecants : { 2, 10, 14, 20 }
Off point 36 = $P_{37} = (2, 1, 1)$ lies on 0 bisecants : { }
Off point 37 = $P_{38} = (3, 1, 1)$ lies on 0 bisecants : { }
Off point 38 = $P_{39} = (4, 1, 1)$ lies on 2 bisecants : { 6, 17 }
Off point 39 = $P_{40} = (5, 1, 1)$ lies on 2 bisecants : { 25, 26 }
Off point 40 = $P_{41} = (6, 1, 1)$ lies on 0 bisecants : { }
Off point 41 = $P_{42} = (7, 1, 1)$ lies on 3 bisecants : { 0, 8, 19 }
Off point 42 = $P_{43} = (8, 1, 1)$ lies on 0 bisecants : { }
Off point 43 = $P_{44} = (9, 1, 1)$ lies on 3 bisecants : { 4, 11, 21 }
Off point 44 = $P_{45} = (10, 1, 1)$ lies on 0 bisecants : { }
Off point 45 = $P_{46} = (11, 1, 1)$ lies on 3 bisecants : { 9, 18, 27 }
Off point 46 = $P_{47} = (12, 1, 1)$ lies on 3 bisecants : { 15, 16, 22 }

Off point 47 = $P_{48} = (13, 1, 1)$ lies on 0 bisecants : { }
Off point 48 = $P_{49} = (14, 1, 1)$ lies on 2 bisecants : { 5, 24 }
Off point 49 = $P_{50} = (15, 1, 1)$ lies on 2 bisecants : { 1, 3 }
Off point 50 = $P_{51} = (16, 1, 1)$ lies on 2 bisecants : { 13, 23 }
Off point 51 = $P_{52} = (0, 2, 1)$ lies on 2 bisecants : { 2, 4 }
Off point 52 = $P_{53} = (1, 2, 1)$ lies on 2 bisecants : { 13, 24 }
Off point 53 = $P_{54} = (2, 2, 1)$ lies on 2 bisecants : { 1, 25 }
Off point 54 = $P_{55} = (3, 2, 1)$ lies on 2 bisecants : { 23, 27 }
Off point 55 = $P_{56} = (4, 2, 1)$ lies on 0 bisecants : { }
Off point 56 = $P_{57} = (5, 2, 1)$ lies on 0 bisecants : { }
Off point 57 = $P_{58} = (6, 2, 1)$ lies on 2 bisecants : { 3, 8 }
Off point 58 = $P_{59} = (7, 2, 1)$ lies on 0 bisecants : { }
Off point 59 = $P_{60} = (8, 2, 1)$ lies on 2 bisecants : { 6, 19 }
Off point 60 = $P_{61} = (9, 2, 1)$ lies on 2 bisecants : { 21, 22 }
Off point 61 = $P_{62} = (10, 2, 1)$ lies on 2 bisecants : { 16, 20 }
Off point 62 = $P_{63} = (11, 2, 1)$ lies on 0 bisecants : { }
Off point 63 = $P_{64} = (12, 2, 1)$ lies on 2 bisecants : { 11, 18 }
Off point 64 = $P_{65} = (13, 2, 1)$ lies on 0 bisecants : { }
Off point 65 = $P_{66} = (14, 2, 1)$ lies on 4 bisecants : { 5, 9, 15, 17 }
Off point 66 = $P_{67} = (15, 2, 1)$ lies on 2 bisecants : { 14, 26 }
Off point 67 = $P_{68} = (16, 2, 1)$ lies on 2 bisecants : { 0, 10 }
Off point 68 = $P_{69} = (0, 3, 1)$ lies on 2 bisecants : { 2, 9 }
Off point 69 = $P_{70} = (1, 3, 1)$ lies on 0 bisecants : { }
Off point 70 = $P_{71} = (2, 3, 1)$ lies on 0 bisecants : { }
Off point 71 = $P_{72} = (3, 3, 1)$ lies on 2 bisecants : { 13, 20 }
Off point 72 = $P_{73} = (4, 3, 1)$ lies on 0 bisecants : { }
Off point 73 = $P_{74} = (5, 3, 1)$ lies on 2 bisecants : { 8, 24 }
Off point 74 = $P_{75} = (6, 3, 1)$ lies on 2 bisecants : { 1, 22 }
Off point 75 = $P_{76} = (7, 3, 1)$ lies on 2 bisecants : { 17, 23 }
Off point 76 = $P_{77} = (8, 3, 1)$ lies on 4 bisecants : { 0, 4, 16, 26 }
Off point 77 = $P_{78} = (9, 3, 1)$ lies on 2 bisecants : { 19, 21 }
Off point 78 = $P_{79} = (10, 3, 1)$ lies on 0 bisecants : { }
Off point 79 = $P_{80} = (11, 3, 1)$ lies on 0 bisecants : { }
Off point 80 = $P_{81} = (12, 3, 1)$ lies on 2 bisecants : { 6, 27 }
Off point 81 = $P_{82} = (13, 3, 1)$ lies on 2 bisecants : { 14, 18 }
Off point 82 = $P_{83} = (14, 3, 1)$ lies on 2 bisecants : { 3, 5 }
Off point 83 = $P_{84} = (15, 3, 1)$ lies on 2 bisecants : { 10, 11 }
Off point 84 = $P_{85} = (16, 3, 1)$ lies on 2 bisecants : { 15, 25 }
Off point 85 = $P_{86} = (0, 4, 1)$ lies on 3 bisecants : { 0, 2, 17 }
Off point 86 = $P_{87} = (1, 4, 1)$ lies on 3 bisecants : { 11, 15, 26 }
Off point 87 = $P_{88} = (2, 4, 1)$ lies on 0 bisecants : { }
Off point 88 = $P_{89} = (3, 4, 1)$ lies on 2 bisecants : { 9, 22 }
Off point 89 = $P_{90} = (4, 4, 1)$ lies on 2 bisecants : { 8, 27 }
Off point 90 = $P_{91} = (5, 4, 1)$ lies on 2 bisecants : { 3, 13 }
Off point 91 = $P_{93} = (7, 4, 1)$ lies on 0 bisecants : { }
Off point 92 = $P_{94} = (8, 4, 1)$ lies on 0 bisecants : { }

Off point 93 = $P_{95} = (9,4,1)$ lies on 2 bisecants : { 21, 24 }
Off point 94 = $P_{96} = (10,4,1)$ lies on 2 bisecants : { 1, 19 }
Off point 95 = $P_{97} = (11,4,1)$ lies on 2 bisecants : { 14, 23 }
Off point 96 = $P_{98} = (12,4,1)$ lies on 0 bisecants : { }
Off point 97 = $P_{99} = (13,4,1)$ lies on 2 bisecants : { 20, 25 }
Off point 98 = $P_{100} = (14,4,1)$ lies on 3 bisecants : { 5, 10, 18 }
Off point 99 = $P_{101} = (15,4,1)$ lies on 0 bisecants : { }
Off point 100 = $P_{102} = (16,4,1)$ lies on 2 bisecants : { 4, 6 }
Off point 101 = $P_{103} = (0,5,1)$ lies on 2 bisecants : { 2, 22 }
Off point 102 = $P_{104} = (1,5,1)$ lies on 0 bisecants : { }
Off point 103 = $P_{105} = (2,5,1)$ lies on 0 bisecants : { }
Off point 104 = $P_{106} = (3,5,1)$ lies on 3 bisecants : { 6, 8, 15 }
Off point 105 = $P_{107} = (4,5,1)$ lies on 2 bisecants : { 11, 16 }
Off point 106 = $P_{108} = (5,5,1)$ lies on 0 bisecants : { }
Off point 107 = $P_{109} = (6,5,1)$ lies on 2 bisecants : { 9, 20 }
Off point 108 = $P_{110} = (7,5,1)$ lies on 2 bisecants : { 4, 13 }
Off point 109 = $P_{111} = (8,5,1)$ lies on 0 bisecants : { }
Off point 110 = $P_{112} = (9,5,1)$ lies on 3 bisecants : { 0, 14, 21 }
Off point 111 = $P_{113} = (10,5,1)$ lies on 2 bisecants : { 17, 25 }
Off point 112 = $P_{114} = (11,5,1)$ lies on 2 bisecants : { 19, 26 }
Off point 113 = $P_{115} = (12,5,1)$ lies on 0 bisecants : { }
Off point 114 = $P_{116} = (13,5,1)$ lies on 4 bisecants : { 3, 10, 24, 27 }
Off point 115 = $P_{117} = (14,5,1)$ lies on 2 bisecants : { 1, 5 }
Off point 116 = $P_{118} = (15,5,1)$ lies on 2 bisecants : { 18, 23 }
Off point 117 = $P_{119} = (16,5,1)$ lies on 0 bisecants : { }
Off point 118 = $P_{120} = (0,6,1)$ lies on 2 bisecants : { 2, 24 }
Off point 119 = $P_{121} = (1,6,1)$ lies on 2 bisecants : { 0, 1 }
Off point 120 = $P_{122} = (2,6,1)$ lies on 3 bisecants : { 8, 16, 23 }
Off point 121 = $P_{124} = (4,6,1)$ lies on 2 bisecants : { 3, 26 }
Off point 122 = $P_{125} = (5,6,1)$ lies on 2 bisecants : { 15, 27 }
Off point 123 = $P_{126} = (6,6,1)$ lies on 0 bisecants : { }
Off point 124 = $P_{127} = (7,6,1)$ lies on 4 bisecants : { 6, 11, 14, 25 }
Off point 125 = $P_{128} = (8,6,1)$ lies on 0 bisecants : { }
Off point 126 = $P_{129} = (9,6,1)$ lies on 3 bisecants : { 9, 13, 21 }
Off point 127 = $P_{130} = (10,6,1)$ lies on 0 bisecants : { }
Off point 128 = $P_{131} = (11,6,1)$ lies on 0 bisecants : { }
Off point 129 = $P_{132} = (12,6,1)$ lies on 2 bisecants : { 10, 19 }
Off point 130 = $P_{133} = (13,6,1)$ lies on 0 bisecants : { }
Off point 131 = $P_{134} = (14,6,1)$ lies on 2 bisecants : { 5, 22 }
Off point 132 = $P_{136} = (16,6,1)$ lies on 2 bisecants : { 18, 20 }
Off point 133 = $P_{137} = (0,7,1)$ lies on 3 bisecants : { 2, 16, 18 }
Off point 134 = $P_{139} = (2,7,1)$ lies on 0 bisecants : { }
Off point 135 = $P_{140} = (3,7,1)$ lies on 0 bisecants : { }
Off point 136 = $P_{141} = (4,7,1)$ lies on 2 bisecants : { 24, 25 }
Off point 137 = $P_{142} = (5,7,1)$ lies on 2 bisecants : { 1, 14 }
Off point 138 = $P_{143} = (6,7,1)$ lies on 2 bisecants : { 4, 23 }

Off point 139 = $P_{145} = (8, 7, 1)$ lies on 0 bisecants : { }
Off point 140 = $P_{146} = (9, 7, 1)$ lies on 2 bisecants : { 20, 21 }
Off point 141 = $P_{147} = (10, 7, 1)$ lies on 2 bisecants : { 0, 11 }
Off point 142 = $P_{148} = (11, 7, 1)$ lies on 4 bisecants : { 6, 10, 13, 22 }
Off point 143 = $P_{149} = (12, 7, 1)$ lies on 2 bisecants : { 3, 9 }
Off point 144 = $P_{150} = (13, 7, 1)$ lies on 2 bisecants : { 17, 19 }
Off point 145 = $P_{151} = (14, 7, 1)$ lies on 3 bisecants : { 5, 26, 27 }
Off point 146 = $P_{152} = (15, 7, 1)$ lies on 0 bisecants : { }
Off point 147 = $P_{153} = (16, 7, 1)$ lies on 0 bisecants : { }
Off point 148 = $P_{154} = (0, 8, 1)$ lies on 2 bisecants : { 2, 8 }
Off point 149 = $P_{155} = (1, 8, 1)$ lies on 2 bisecants : { 18, 25 }
Off point 150 = $P_{156} = (2, 8, 1)$ lies on 2 bisecants : { 0, 20 }
Off point 151 = $P_{157} = (3, 8, 1)$ lies on 2 bisecants : { 3, 14 }
Off point 152 = $P_{158} = (4, 8, 1)$ lies on 0 bisecants : { }
Off point 153 = $P_{159} = (5, 8, 1)$ lies on 0 bisecants : { }
Off point 154 = $P_{160} = (6, 8, 1)$ lies on 2 bisecants : { 17, 27 }
Off point 155 = $P_{162} = (8, 8, 1)$ lies on 2 bisecants : { 22, 24 }
Off point 156 = $P_{163} = (9, 8, 1)$ lies on 3 bisecants : { 1, 15, 21 }
Off point 157 = $P_{164} = (10, 8, 1)$ lies on 2 bisecants : { 10, 23 }
Off point 158 = $P_{165} = (11, 8, 1)$ lies on 0 bisecants : { }
Off point 159 = $P_{166} = (12, 8, 1)$ lies on 0 bisecants : { }
Off point 160 = $P_{167} = (13, 8, 1)$ lies on 2 bisecants : { 11, 13 }
Off point 161 = $P_{168} = (14, 8, 1)$ lies on 3 bisecants : { 4, 5, 19 }
Off point 162 = $P_{169} = (15, 8, 1)$ lies on 3 bisecants : { 6, 9, 16 }
Off point 163 = $P_{170} = (16, 8, 1)$ lies on 0 bisecants : { }
Off point 164 = $P_{171} = (0, 9, 1)$ lies on 2 bisecants : { 2, 26 }
Off point 165 = $P_{172} = (1, 9, 1)$ lies on 2 bisecants : { 9, 14 }
Off point 166 = $P_{173} = (2, 9, 1)$ lies on 2 bisecants : { 6, 18 }
Off point 167 = $P_{174} = (3, 9, 1)$ lies on 0 bisecants : { }
Off point 168 = $P_{175} = (4, 9, 1)$ lies on 0 bisecants : { }
Off point 169 = $P_{176} = (5, 9, 1)$ lies on 2 bisecants : { 4, 22 }
Off point 170 = $P_{177} = (6, 9, 1)$ lies on 0 bisecants : { }
Off point 171 = $P_{178} = (7, 9, 1)$ lies on 0 bisecants : { }
Off point 172 = $P_{179} = (8, 9, 1)$ lies on 0 bisecants : { }
Off point 173 = $P_{180} = (9, 9, 1)$ lies on 2 bisecants : { 10, 21 }
Off point 174 = $P_{181} = (10, 9, 1)$ lies on 0 bisecants : { }
Off point 175 = $P_{182} = (11, 9, 1)$ lies on 3 bisecants : { 0, 3, 15 }
Off point 176 = $P_{183} = (12, 9, 1)$ lies on 2 bisecants : { 20, 24 }
Off point 177 = $P_{184} = (13, 9, 1)$ lies on 2 bisecants : { 1, 16 }
Off point 178 = $P_{185} = (14, 9, 1)$ lies on 2 bisecants : { 5, 23 }
Off point 179 = $P_{186} = (15, 9, 1)$ lies on 4 bisecants : { 13, 19, 25, 27 }
Off point 180 = $P_{187} = (16, 9, 1)$ lies on 3 bisecants : { 8, 11, 17 }
Off point 181 = $P_{188} = (0, 10, 1)$ lies on 4 bisecants : { 1, 2, 12, 13 }
Off point 182 = $P_{189} = (1, 10, 1)$ lies on 2 bisecants : { 12, 23 }
Off point 183 = $P_{190} = (2, 10, 1)$ lies on 4 bisecants : { 3, 11, 12, 22 }
Off point 184 = $P_{191} = (3, 10, 1)$ lies on 3 bisecants : { 0, 12, 18 }

Off point 185 = $P_{192} = (4, 10, 1)$ lies on 2 bisecants : { 9, 12 }
Off point 186 = $P_{193} = (5, 10, 1)$ lies on 2 bisecants : { 12, 20 }
Off point 187 = $P_{194} = (6, 10, 1)$ lies on 2 bisecants : { 6, 12 }
Off point 188 = $P_{195} = (7, 10, 1)$ lies on 2 bisecants : { 12, 27 }
Off point 189 = $P_{196} = (8, 10, 1)$ lies on 2 bisecants : { 10, 12 }
Off point 190 = $P_{197} = (9, 10, 1)$ lies on 3 bisecants : { 12, 17, 21 }
Off point 191 = $P_{198} = (10, 10, 1)$ lies on 2 bisecants : { 12, 26 }
Off point 192 = $P_{199} = (11, 10, 1)$ lies on 2 bisecants : { 12, 16 }
Off point 193 = $P_{200} = (12, 10, 1)$ lies on 2 bisecants : { 12, 25 }
Off point 194 = $P_{201} = (13, 10, 1)$ lies on 3 bisecants : { 4, 12, 15 }
Off point 195 = $P_{202} = (14, 10, 1)$ lies on 2 bisecants : { 5, 12 }
Off point 196 = $P_{203} = (15, 10, 1)$ lies on 2 bisecants : { 8, 12 }
Off point 197 = $P_{204} = (16, 10, 1)$ lies on 4 bisecants : { 12, 14, 19, 24 }
Off point 198 = $P_{205} = (0, 11, 1)$ lies on 2 bisecants : { 2, 19 }
Off point 199 = $P_{206} = (1, 11, 1)$ lies on 0 bisecants : { }
Off point 200 = $P_{207} = (2, 11, 1)$ lies on 2 bisecants : { 13, 17 }
Off point 201 = $P_{208} = (3, 11, 1)$ lies on 2 bisecants : { 24, 26 }
Off point 202 = $P_{209} = (4, 11, 1)$ lies on 3 bisecants : { 1, 4, 18 }
Off point 203 = $P_{210} = (5, 11, 1)$ lies on 2 bisecants : { 11, 23 }
Off point 204 = $P_{211} = (6, 11, 1)$ lies on 0 bisecants : { }
Off point 205 = $P_{212} = (7, 11, 1)$ lies on 2 bisecants : { 9, 10 }
Off point 206 = $P_{213} = (8, 11, 1)$ lies on 0 bisecants : { }
Off point 207 = $P_{214} = (9, 11, 1)$ lies on 3 bisecants : { 16, 21, 25 }
Off point 208 = $P_{215} = (10, 11, 1)$ lies on 2 bisecants : { 3, 6 }
Off point 209 = $P_{216} = (11, 11, 1)$ lies on 0 bisecants : { }
Off point 210 = $P_{218} = (13, 11, 1)$ lies on 0 bisecants : { }
Off point 211 = $P_{219} = (14, 11, 1)$ lies on 3 bisecants : { 5, 8, 14 }
Off point 212 = $P_{220} = (15, 11, 1)$ lies on 2 bisecants : { 15, 20 }
Off point 213 = $P_{221} = (16, 11, 1)$ lies on 2 bisecants : { 22, 27 }
Off point 214 = $P_{222} = (0, 12, 1)$ lies on 2 bisecants : { 2, 15 }
Off point 215 = $P_{223} = (1, 12, 1)$ lies on 2 bisecants : { 3, 19 }
Off point 216 = $P_{224} = (2, 12, 1)$ lies on 0 bisecants : { }
Off point 217 = $P_{225} = (3, 12, 1)$ lies on 0 bisecants : { }
Off point 218 = $P_{226} = (4, 12, 1)$ lies on 2 bisecants : { 0, 13 }
Off point 219 = $P_{228} = (6, 12, 1)$ lies on 2 bisecants : { 10, 25 }
Off point 220 = $P_{229} = (7, 12, 1)$ lies on 2 bisecants : { 16, 24 }
Off point 221 = $P_{230} = (8, 12, 1)$ lies on 4 bisecants : { 1, 11, 20, 27 }
Off point 222 = $P_{231} = (9, 12, 1)$ lies on 2 bisecants : { 21, 23 }
Off point 223 = $P_{233} = (11, 12, 1)$ lies on 0 bisecants : { }
Off point 224 = $P_{234} = (12, 12, 1)$ lies on 3 bisecants : { 4, 14, 17 }
Off point 225 = $P_{235} = (13, 12, 1)$ lies on 3 bisecants : { 8, 22, 26 }
Off point 226 = $P_{236} = (14, 12, 1)$ lies on 2 bisecants : { 5, 6 }
Off point 227 = $P_{237} = (15, 12, 1)$ lies on 0 bisecants : { }
Off point 228 = $P_{238} = (16, 12, 1)$ lies on 0 bisecants : { }
Off point 229 = $P_{239} = (0, 13, 1)$ lies on 2 bisecants : { 2, 27 }
Off point 230 = $P_{240} = (1, 13, 1)$ lies on 2 bisecants : { 6, 20 }

Off point 231 = $P_{241} = (2, 13, 1)$ lies on 2 bisecants : { 15, 19 }
Off point 232 = $P_{242} = (3, 13, 1)$ lies on 2 bisecants : { 4, 25 }
Off point 233 = $P_{243} = (4, 13, 1)$ lies on 0 bisecants : { }
Off point 234 = $P_{244} = (5, 13, 1)$ lies on 3 bisecants : { 10, 16, 17 }
Off point 235 = $P_{245} = (6, 13, 1)$ lies on 3 bisecants : { 13, 18, 26 }
Off point 236 = $P_{246} = (7, 13, 1)$ lies on 0 bisecants : { }
Off point 237 = $P_{247} = (8, 13, 1)$ lies on 0 bisecants : { }
Off point 238 = $P_{248} = (9, 13, 1)$ lies on 2 bisecants : { 3, 21 }
Off point 239 = $P_{249} = (10, 13, 1)$ lies on 2 bisecants : { 14, 22 }
Off point 240 = $P_{250} = (11, 13, 1)$ lies on 2 bisecants : { 11, 24 }
Off point 241 = $P_{251} = (12, 13, 1)$ lies on 2 bisecants : { 1, 8 }
Off point 242 = $P_{252} = (13, 13, 1)$ lies on 3 bisecants : { 0, 9, 23 }
Off point 243 = $P_{254} = (15, 13, 1)$ lies on 0 bisecants : { }
Off point 244 = $P_{255} = (16, 13, 1)$ lies on 0 bisecants : { }
Off point 245 = $P_{256} = (0, 14, 1)$ lies on 4 bisecants : { 2, 3, 23, 25 }
Off point 246 = $P_{257} = (1, 14, 1)$ lies on 0 bisecants : { }
Off point 247 = $P_{258} = (2, 14, 1)$ lies on 0 bisecants : { }
Off point 248 = $P_{259} = (3, 14, 1)$ lies on 2 bisecants : { 16, 19 }
Off point 249 = $P_{260} = (4, 14, 1)$ lies on 2 bisecants : { 10, 15 }
Off point 250 = $P_{261} = (5, 14, 1)$ lies on 2 bisecants : { 0, 6 }
Off point 251 = $P_{262} = (6, 14, 1)$ lies on 0 bisecants : { }
Off point 252 = $P_{263} = (7, 14, 1)$ lies on 2 bisecants : { 18, 22 }
Off point 253 = $P_{264} = (8, 14, 1)$ lies on 2 bisecants : { 13, 14 }
Off point 254 = $P_{265} = (9, 14, 1)$ lies on 2 bisecants : { 21, 27 }
Off point 255 = $P_{266} = (10, 14, 1)$ lies on 0 bisecants : { }
Off point 256 = $P_{267} = (11, 14, 1)$ lies on 3 bisecants : { 4, 8, 20 }
Off point 257 = $P_{268} = (12, 14, 1)$ lies on 0 bisecants : { }
Off point 258 = $P_{269} = (13, 14, 1)$ lies on 0 bisecants : { }
Off point 259 = $P_{270} = (14, 14, 1)$ lies on 2 bisecants : { 5, 11 }
Off point 260 = $P_{271} = (15, 14, 1)$ lies on 2 bisecants : { 17, 24 }
Off point 261 = $P_{272} = (16, 14, 1)$ lies on 3 bisecants : { 1, 9, 26 }
Off point 262 = $P_{273} = (0, 15, 1)$ lies on 2 bisecants : { 2, 11 }
Off point 263 = $P_{274} = (1, 15, 1)$ lies on 2 bisecants : { 16, 27 }
Off point 264 = $P_{275} = (2, 15, 1)$ lies on 3 bisecants : { 4, 9, 24 }
Off point 265 = $P_{276} = (3, 15, 1)$ lies on 2 bisecants : { 1, 10 }
Off point 266 = $P_{277} = (4, 15, 1)$ lies on 4 bisecants : { 19, 20, 22, 23 }
Off point 267 = $P_{278} = (5, 15, 1)$ lies on 0 bisecants : { }
Off point 268 = $P_{279} = (6, 15, 1)$ lies on 2 bisecants : { 14, 15 }
Off point 269 = $P_{280} = (7, 15, 1)$ lies on 0 bisecants : { }
Off point 270 = $P_{281} = (8, 15, 1)$ lies on 3 bisecants : { 3, 17, 18 }
Off point 271 = $P_{282} = (9, 15, 1)$ lies on 3 bisecants : { 6, 21, 26 }
Off point 272 = $P_{283} = (10, 15, 1)$ lies on 2 bisecants : { 8, 13 }
Off point 273 = $P_{284} = (11, 15, 1)$ lies on 0 bisecants : { }
Off point 274 = $P_{285} = (12, 15, 1)$ lies on 0 bisecants : { }
Off point 275 = $P_{286} = (13, 15, 1)$ lies on 0 bisecants : { }
Off point 276 = $P_{287} = (14, 15, 1)$ lies on 3 bisecants : { 0, 5, 25 }

Off point 277 = $P_{288} = (15, 15, 1)$ lies on 0 bisecants : { }
Off point 278 = $P_{289} = (16, 15, 1)$ lies on 0 bisecants : { }
Off point 279 = $P_{290} = (0, 16, 1)$ lies on 2 bisecants : { 2, 7 }
Off point 280 = $P_{291} = (1, 16, 1)$ lies on 3 bisecants : { 7, 17, 22 }
Off point 281 = $P_{292} = (2, 16, 1)$ lies on 3 bisecants : { 7, 10, 26 }
Off point 282 = $P_{293} = (3, 16, 1)$ lies on 2 bisecants : { 7, 11 }
Off point 283 = $P_{294} = (4, 16, 1)$ lies on 2 bisecants : { 7, 14 }
Off point 284 = $P_{295} = (5, 16, 1)$ lies on 3 bisecants : { 7, 9, 19 }
Off point 285 = $P_{296} = (6, 16, 1)$ lies on 3 bisecants : { 0, 7, 24 }
Off point 286 = $P_{297} = (7, 16, 1)$ lies on 2 bisecants : { 1, 7 }
Off point 287 = $P_{298} = (8, 16, 1)$ lies on 3 bisecants : { 7, 15, 23 }
Off point 288 = $P_{299} = (9, 16, 1)$ lies on 4 bisecants : { 7, 8, 18, 21 }
Off point 289 = $P_{300} = (10, 16, 1)$ lies on 3 bisecants : { 4, 7, 27 }
Off point 290 = $P_{301} = (11, 16, 1)$ lies on 2 bisecants : { 7, 25 }
Off point 291 = $P_{302} = (12, 16, 1)$ lies on 2 bisecants : { 7, 13 }
Off point 292 = $P_{303} = (13, 16, 1)$ lies on 2 bisecants : { 6, 7 }
Off point 293 = $P_{304} = (14, 16, 1)$ lies on 3 bisecants : { 5, 7, 20 }
Off point 294 = $P_{306} = (16, 16, 1)$ lies on 3 bisecants : { 3, 7, 16 }

The gradient

The gradient of the quartic curve is :

$$13X_0^3 + 11X_1^3 + 11X_2^3 + 6X_0^2X_1 + 16X_0^2X_2 + 10X_0X_1^2 + 2X_1^2X_2 + 4X_0X_2^2 \\ +11X_0X_1X_2$$

$$(13, 11, 11, 6, 16, 10, 2, 4, 0, 11)$$

$$2X_0^3 + X_1^3 + 9X_2^3 + 10X_0^2X_1 + 14X_0^2X_2 + 16X_0X_1^2 + 9X_1^2X_2 + 4X_1X_2^2 \\ +4X_0X_1X_2$$

$$(2, 1, 9, 10, 14, 16, 9, 0, 4, 4)$$

$$11X_0^3 + 3X_1^3 + 9X_2^3 + 14X_0^2X_1 + 4X_0^2X_2 + 2X_0X_1^2 + 4X_1^2X_2 + 16X_0X_2^2 \\ +10X_1X_2^2$$

$$(11, 3, 9, 14, 4, 2, 4, 16, 10, 0)$$